THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET B

Chapter 13: **Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the insolation is 1 kW m^{-2}, the solar energy received by 2 m^2 in 3 h is: 1

(a) 9 kWh **(b)** 5 kWh **(c)** 6 kWh **(d)** 12 kWh

2. The solar constant is about 1400 J s^{-1} m^{-2}. The energy arriving on 5 m^2 (perpendicular to the Sun's rays) in 1 s is: 1

(a) 7000 J **(b)** 8000 J **(c)** 14000 J **(d)** 3500 J

3. A wavelength of 400 nm in metres is: 1

(a) 400 × 10^{-3} m **(b)** 400 × 10^{-9} m **(c)** 400 × 10^{-6} m **(d)** 400 × 10^{-12} m

4. Electromagnetic waves: 1

(a) are slower than sound **(b)** are longitudinal only
(c) need a medium **(d)** can travel through vacuum

5. The amount of solar radiation reaching the Earth's surface is called: 1

(a) albedo **(b)** insolation **(c)** solar constant **(d)** transpiration

6. Which gas is the most abundant in the Earth's atmosphere? 1

(a) carbon dioxide **(b)** argon **(c)** nitrogen **(d)** oxygen

7. The greenhouse effect: 1

(a) makes the Earth colder **(b)** is entirely harmful
(c) keeps the Earth warm enough for life **(d)** occurs only in greenhouses

8. Winds are caused by: 1

(a) tides only **(b)** volcanic eruptions
(c) rotation of the Moon **(d)** uneven heating of the Earth's surface

9. Albedo is the: 1

(a) temperature of the air
(b) pressure of the atmosphere
(c) fraction of incoming solar radiation reflected by a surface
(d) energy absorbed by plants

10. The largest reservoir of carbon on Earth is the: 1

(a) living organisms **(b)** oceans **(c)** soil only **(d)** atmosphere

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. How much solar energy falls on 1 m^2 in one hour at an insolation of 1 kW m^{-2}? 2

12. Why do polar regions receive less heat than the equator? 2

13. What is the greenhouse effect? 2

14. Name two greenhouse gases and a human source for each. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain how the Earth's tilt and shape cause seasons and uneven heating. 3

16. Describe the layers of the atmosphere with one feature each. 3

17. State two roles of the atmosphere in supporting life. 3

OR

Why is Venus hotter than Mercury although Mercury is nearer to the Sun? 3

18. Explain how the ozone hole formed and how it is recovering. 3

OR

Explain land and sea breezes. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Why is the heating of the Earth uneven? 5
(ii) How does it drive winds?
(iii) Explain land and sea breezes.

OR

(i) Describe the structure of the atmosphere. 5
(ii) What is its composition?
(iii) How does it protect life?

20. **Case study:** The ozone layer thinned over Antarctica in the late 20th century. 5
(i) What caused it?
(ii) What harm does extra UV cause?
(iii) What international action was taken?
(iv) What is the current status?

OR

(i) Explain the greenhouse effect. 5
(ii) What is global warming?
(iii) State four consequences.

END OF QUESTION PAPER — Total: 40 Marks

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